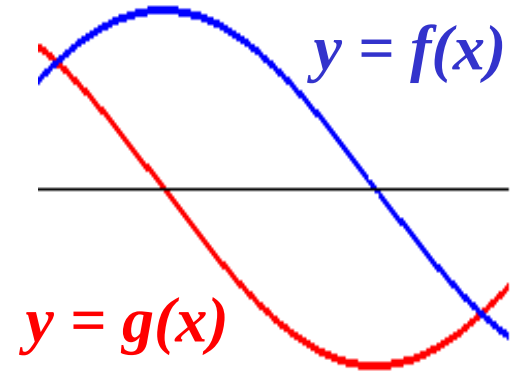


Areas Between Curves

To find the area:

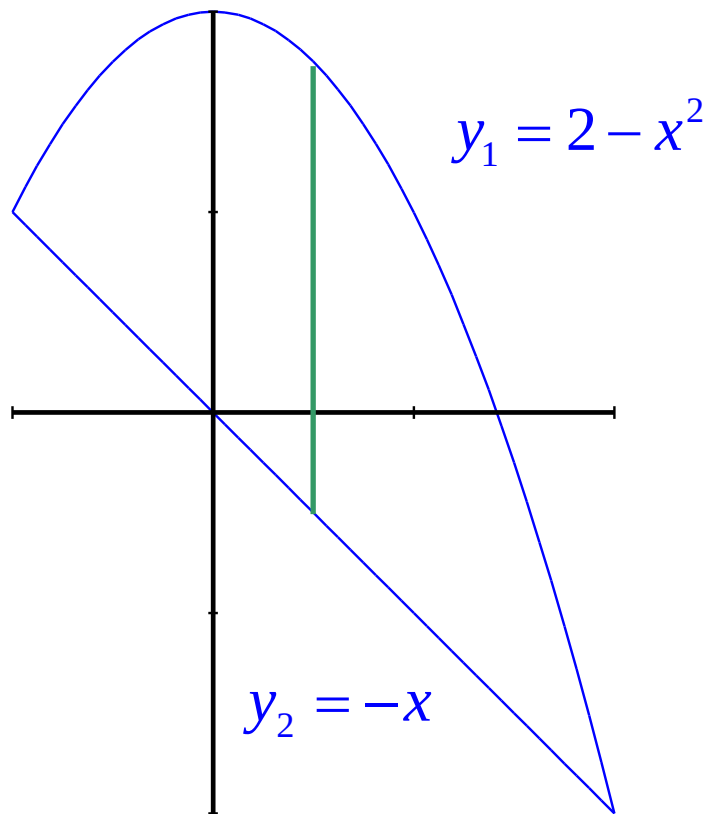
- divide the area into n strips of equal width
- approximate the i th strip by a rectangle with base Δx and height $f(x_i) - g(x_i)$.
- the sum of the rectangle areas is a good approximation
- the approximation is getting better as $n \rightarrow \infty$.



The area A of the region bounded by the curves $y=f(x)$, $y=g(x)$, and the lines $x=a$, $x=b$, where f and g are continuous and $f(x) \geq g(x)$ for all x in $[a,b]$, is

$$A = \int_a^b [f(x) - g(x)] dx$$

Example



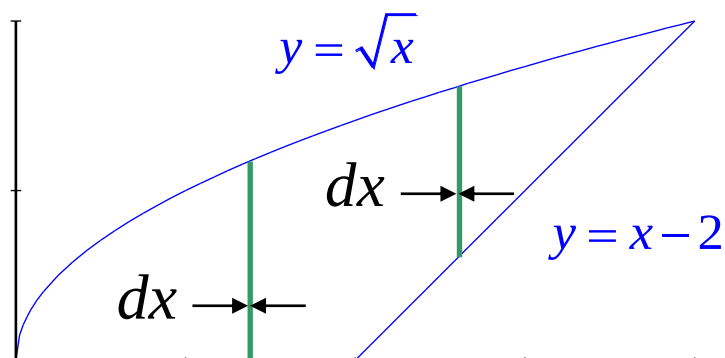
$$\int_{-1}^2 2 - x^2 + x \, dx$$

$$2x - \frac{1}{3}x^3 + \frac{1}{2}x^2 \Big|_{-1}^2$$

$$\left(4 - \frac{8}{3} + 2\right) - \left(-2 + \frac{1}{3} + \frac{1}{2}\right)$$

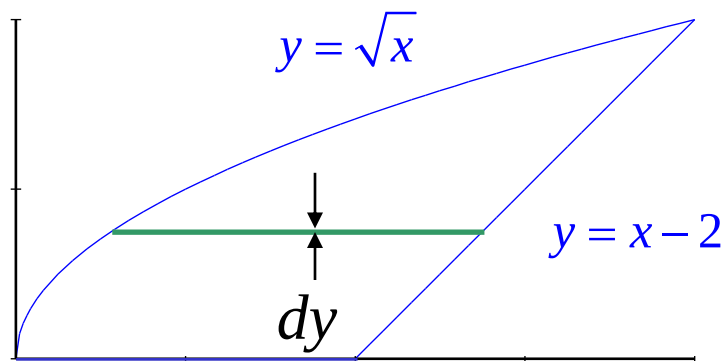
$$6 - \frac{8}{3} + 2 - \frac{1}{3} - \frac{1}{2}$$

$$\frac{36 - 16 + 12 - 2 - 3}{6} = \frac{27}{6} = \frac{9}{2}$$



If we try vertical strips, we have to integrate in two parts:

$$\int_0^2 \sqrt{x} \, dx + \int_2^4 \sqrt{x} - (x - 2) \, dx$$



We can find the same area using a horizontal strip.

Since the width of the strip is dy , we find the length of the strip by solving for x in terms of y .

$$y = \sqrt{x} \quad y = x - 2$$

$$y^2 = x \quad y + 2 = x$$

$$\int_0^2 (y + 2) - y^2 \, dy \quad \left. \frac{1}{2} y^2 + 2y - \frac{1}{3} y^3 \right|_0^2 \quad 2 + 4 - \frac{8}{3} = \frac{10}{3}$$

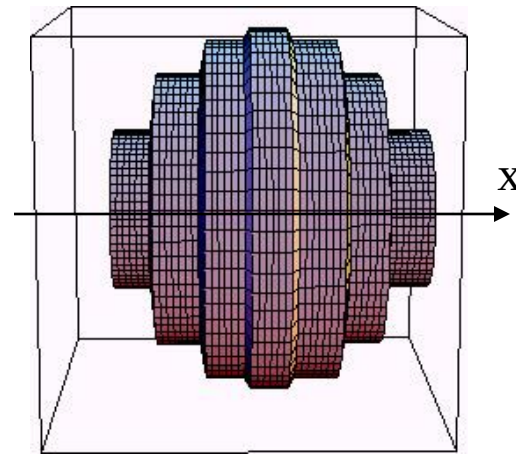
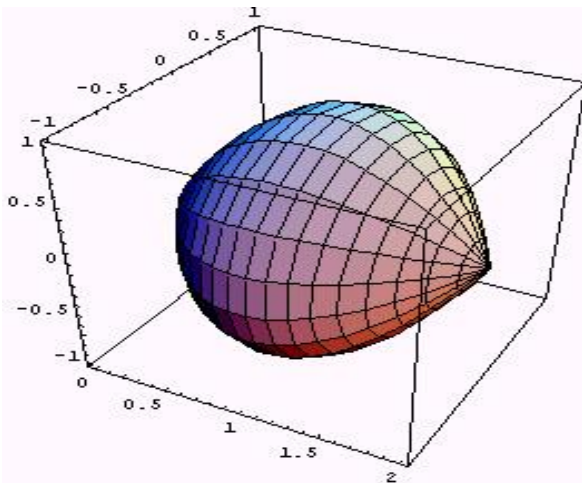
General Strategy for Area Between Curves:

- ① Sketch the curves.
- ② Decide on vertical or horizontal strips. (Pick whichever is easier to write formulas for the length of the strip, and/or whichever will let you integrate fewer times.)
- ③ Write an expression for the area of the strip.
(If the width is dx , the length must be in terms of x .
If the width is dy , the length must be in terms of y .)
- ④ Find the limits of integration. (If using dx , the limits are x values; if using dy , the limits are y values.)
- ⑤ Integrate to find area.

Volumes

To find the volume of a solid S :

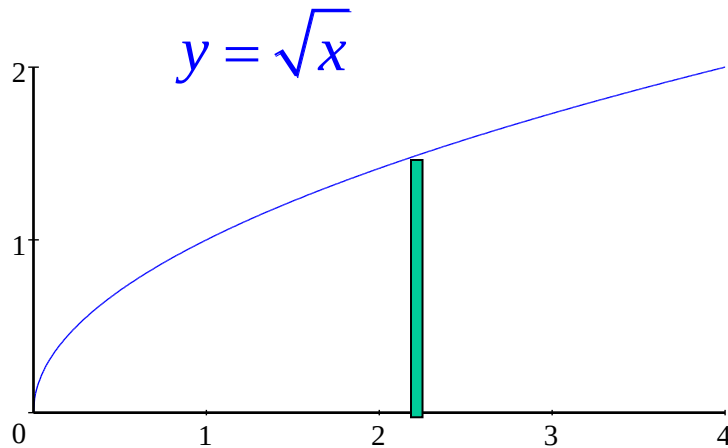
- Divide S into n “slabs” of equal width Δx (think of slicing a loaf of bread)
- Approximate the i th slab by a cylinder with base area $A(x_i)$ and “height” Δx . The volume of the cylinder is $A(x_i)\Delta x$
- the sum of the cylinder areas is a good approximation for the volume of the solid
- the approximation is getting better as $n \rightarrow \infty$.



Let S be a solid that lies between $x=a$ and $x=b$. If the cross-sectional area of S in the plane P_x , perpendicular to the x -axis, is $A(x)$, where A is an integrable function, then the **volume** of S is

$$V = \lim_{\max \Delta x_i \rightarrow 0} \sum_{i=1}^n A(x_i^*) \Delta x_i = \int_a^b A(x) dx$$

Example of a disk



How could we find the volume of the cone?

One way would be to cut it into a series of disks (flat circular cylinders) and add their volumes.

The volume of each disk is:

$$\pi r^2 \cdot \text{the thickness}$$

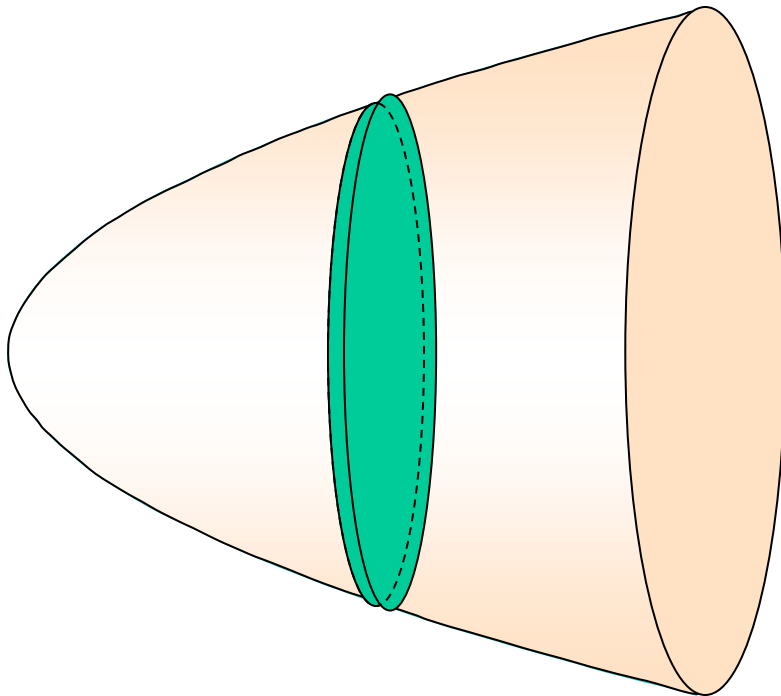
$\pi (\sqrt{x})^2 dx$

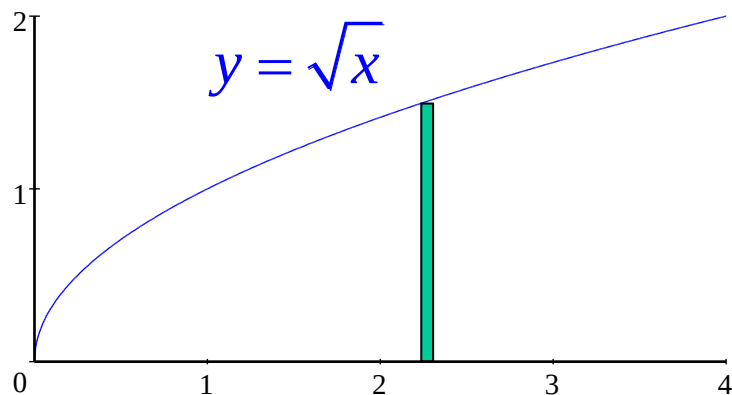
Red arrows point from the text above to the terms in the equation below: π from the first π , r^2 from $(\sqrt{x})^2$, and the thickness from dx .

In this case:

$r =$ the y value of the function

thickness = a small change in $x = dx$





The volume of each flat cylinder (disk) is:

$\pi r^2 \cdot \text{the thickness}$

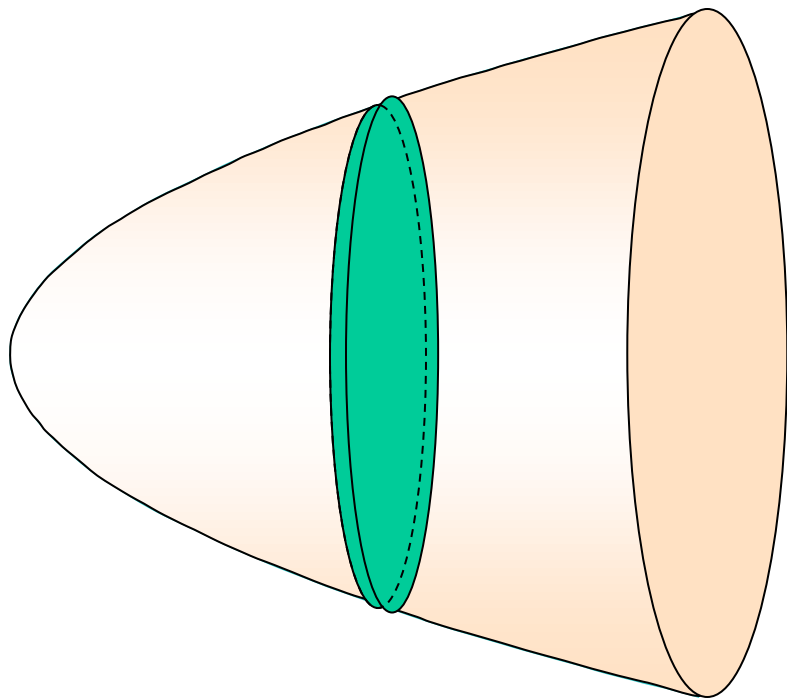
$$\pi (\sqrt{x})^2 dx$$

If we add the volumes, we get:

$$\int_0^4 \pi (\sqrt{x})^2 dx$$

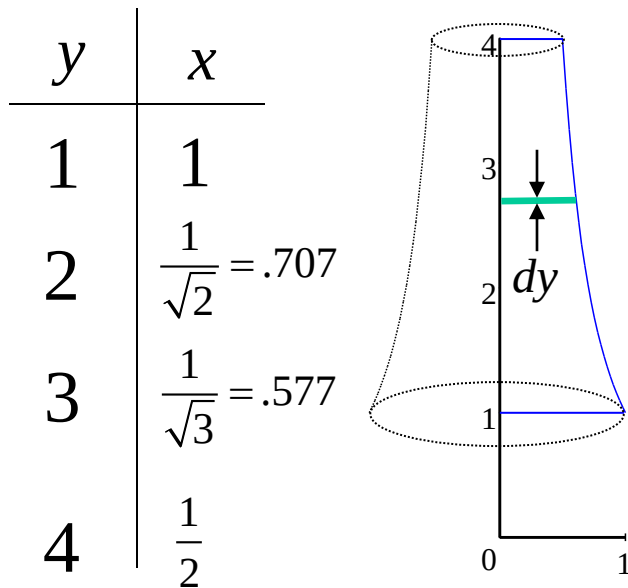
$$= \int_0^4 \pi x dx$$

$$= \frac{\pi}{2} x^2 \Big|_0^4 = 8\pi$$



Example of rotating the region about y -axis

The region between the curve $x = \frac{1}{\sqrt{y}}$, $1 \leq y \leq 4$ and the y -axis is revolved about the y -axis. Find the volume.



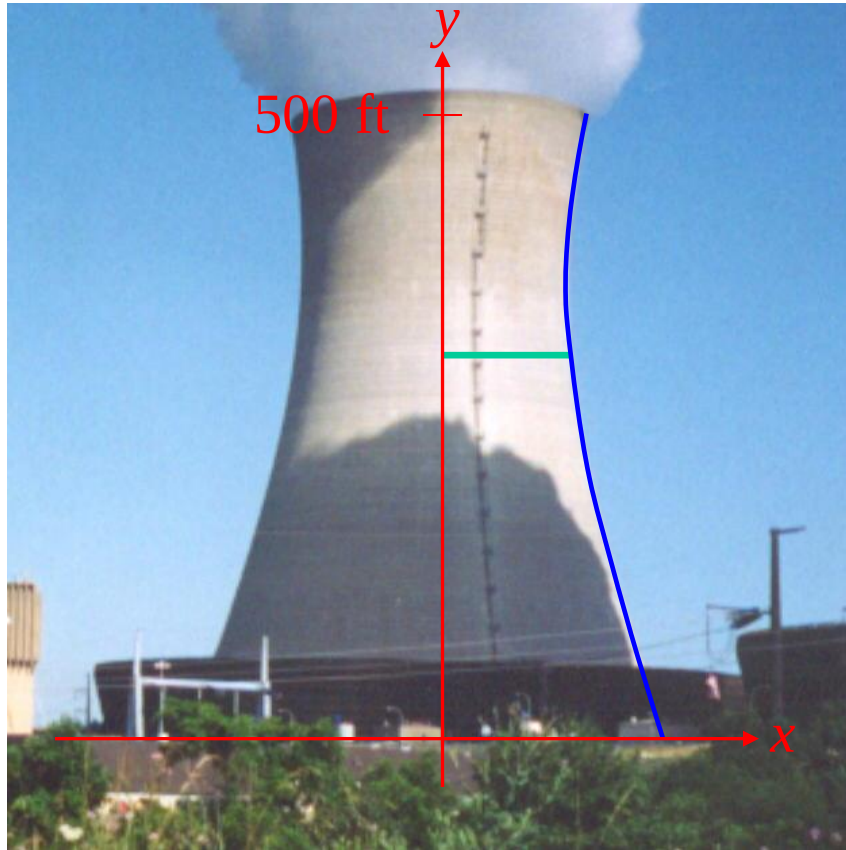
We use a horizontal disk.

The thickness is dy .

The radius is the x value of the function $= \frac{1}{\sqrt{y}}$.

$$V = \int_1^4 \underbrace{\pi \left(\frac{1}{\sqrt{y}} \right)^2}_{\text{volume of disk}} dy = \int_1^4 \pi \frac{1}{y} dy$$

$$= \pi \ln y \Big|_1^4 = \pi (\ln 4 - \cancel{\ln 1}^0) = \pi \ln 2^2 = 2\pi \ln 2$$



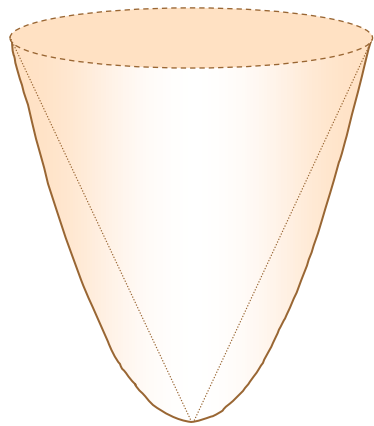
The natural draft cooling tower shown at left is about 500 feet high and its shape can be approximated by the graph of this equation revolved about the y-axis:

$$x = .000574y^2 - .439y + 185$$

The volume can be calculated using the disk method with a horizontal disk.

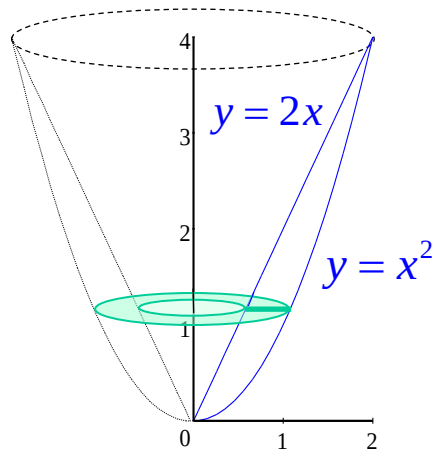
$$\pi \int_0^{500} (.000574y^2 - .439y + 185)^2 dy \approx 24,700,000 \text{ ft}^3$$

Example of a washer



$$y = x^2 \quad y = 2x$$

$$\sqrt{y} = x \quad \frac{y}{2} = x$$



The region bounded by $y = x^2$ and $y = 2x$ is revolved about the y -axis. Find the volume.

If we use a horizontal slice:

The “disk” now has a hole in it, making it a “washer”.

The volume of the washer is: $(\pi R^2 - \pi r^2) \cdot \text{thickness}$

$$V = \int_0^4 \pi \left((\sqrt{y})^2 - \left(\frac{y}{2} \right)^2 \right) dy$$

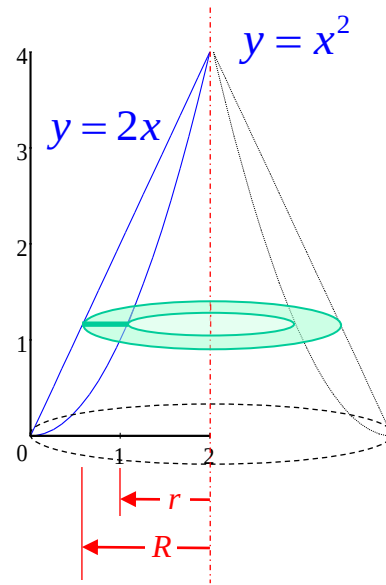
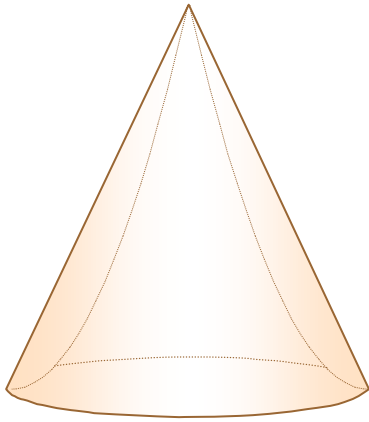
$$V = \int_0^4 \pi \left(y - \frac{1}{4} y^2 \right) dy$$

$$\pi (R^2 - r^2) dy$$

outer radius inner radius

$$V = \pi \int_0^4 y - \frac{1}{4} y^2 dy = \pi \left[\frac{1}{2} y^2 - \frac{1}{12} y^3 \right]_0^4 = \pi \left[8 - \frac{16}{3} \right] = \frac{8\pi}{3}$$





If the same region is rotated about the line $x=2$:

The outer radius is:

$$R = 2 - \frac{y}{2}$$

The inner radius is:

$$r = 2 - \sqrt{y}$$

$$= \pi \int_0^4 \cancel{4} - 2y + \frac{y^2}{4} - \cancel{4} + 4\sqrt{y} - y \, dy$$

$$= \pi \int_0^4 -3y + \frac{1}{4}y^2 + 4y^{\frac{1}{2}} \, dy$$

$$= \pi \cdot \left[-\frac{3}{2}y^2 + \frac{1}{12}y^3 + \frac{8}{3}y^{\frac{3}{2}} \right]_0^4$$

$$= \pi \cdot \left[-24 + \frac{16}{3} + \frac{64}{3} \right] = \frac{8\pi}{3}$$

$$y = x^2$$

$$y = 2x$$

$$\sqrt{y} = x$$

$$\frac{y}{2} = x$$

$$V = \pi \int_0^4 R^2 - r^2 \, dy$$

$$= \pi \int_0^4 \left(2 - \frac{y}{2} \right)^2 - \left(2 - \sqrt{y} \right)^2 \, dy$$

$$= \pi \int_0^4 \left(4 - 2y + \frac{y^2}{4} \right) - \left(4 - 4\sqrt{y} + y \right) \, dy$$

Volumes of Solids of Revolution

The solids we considered are examples of **solids of revolution** because they are obtained by revolving a region about a line. In general, we calculate the volume of a solid of revolution by using the basic defining formula

$$V = \int_a^b A(x)dx \quad \text{or} \quad V = \int_c^d A(y)dy$$

and we find the cross-sectional area $A(x)$ or $A(y)$ in one of the following ways:

- If the cross-section is a **disk**, we find the radius of the disk (in terms of x or y) and use

$$A = \pi(\text{radius})^2$$

- If the cross-section is a **washer**, we find the inner radius r_{in} and outer radius r_{out} and compute the area of the washer by subtracting the area of the inner disk from the area of the outer disk:

$$A = \pi(\text{outer radius})^2 - \pi(\text{inner radius})^2$$